Rishi Garhyan

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Education

University of Illinois Urbana-Champaign — Aug. 2024 – May 2028
Bachelor of Science in Computer Science, Grainger College of Engineering — *GPA: 3.98/4.00*

- Dean's List, all semesters. Relevant coursework: Algorithms & Models of Computation, Data Structures, Computer Architecture, Interactive Computer Graphics

Experience

Software Development Intern, Computer Graphics — May 2026 – Present
Brunswick Corporation — BI Design Lab — *Champaign, IL*

- Primary developer on the UE5.7 (C++) simulator behind Future Helm, Brunswick's flagship CES 2027 marine showcase; reviewed by senior management including C-suite for January public exhibition
- Own the multiprotocol C++ data layer (TCP, UDP, CAN) exchanging vessel state, engine telemetry, and navigation data bidirectionally with a teammate's multi-function display app; extended its existing CAN interface with new signals
- Prototyped two planned Brunswick features: velocity-scaled reverse-thrust collision avoidance from forward-projected volumes, and spline path selection weighted by proximity and user preferences; co-designed the single-integer binary encoding carrying those weights from an unnetworked planning app
- Profiled and optimized simulator runtime against a 60 fps target via level-of-detail tuning, GPU instancing, and a shift from polling to event-driven messaging
- Top-3 finalist for Most Outstanding Undergraduate Student Intern, 2026 UIUC Research Park Intern Awards (program: 800+ interns, 120+ companies)

Virtual Reality Software Developer — Oct. 2025 – Present
Immersive Learning Lab, Grainger College of Engineering — *Champaign, IL*

- Designed and programmed an educational VR application in C# visualizing the University of Bern's MANiaC mass spectrometer, an instrument selected for ESA's Comet Interceptor mission
- Developed physics simulations, live data graphing (XCharts), animation systems, and custom interaction scripts
- Built a near-photorealistic VR environment with physics-driven interactable objects and NavMesh-driven NPCs
- Presented the project at the IMMERSE Annual Symposium (April 2026) via a poster and live demo

Projects

SportsBot | *Python, discord.py, asyncio, REST* — Aug. 2025 – Dec. 2025

- Built an autonomous Discord bot in Python using asyncio for concurrent, scheduled background tasks
- Integrated third-party REST APIs for automated data retrieval and delivery; designed a slash-command user interface

Pelagos | *Unity, C#, Steam — shipped product* — 2022 – 2023

- Self-published an interactive marine-photography application on Steam with 30k+ downloads, adopted into local schools' environmental-science curricula
- Owned the full product lifecycle solo: core systems, UI, asset pipeline, launch, and post-release support

Party School | *Unreal Engine 5, multiplayer* — Feb. 2026 – Present

- Developing a multiplayer sandbox application in a five-person team; built the shared core project base, supporting multiplayer integration and reviewing teammates' code
- Streamed a real-world city-scale map into the engine from the Google Map Tiles API via the BLOSM plugin

Object Detection Benchmark | *Python, Detectron2, YOLOv8* — 2023 – 2024

- Benchmarked Faster R-CNN (Detectron2) against YOLOv8 on the COCO dataset and authored a research paper comparing accuracy and inference-speed tradeoffs

Technical Skills

Languages: C++, Python, C#, Java
Frameworks & Technologies: Unreal Engine 5, Unity, PyTorch, asyncio, REST APIs, TCP/UDP, CAN bus, OpenXR
Tools: Git, Perforce, Blender

Other Involvements

VR Club at UIUC, ACM GameBuilders, Illini Fly Fishing, Illini Metagamers